Feasibility Iteration 2

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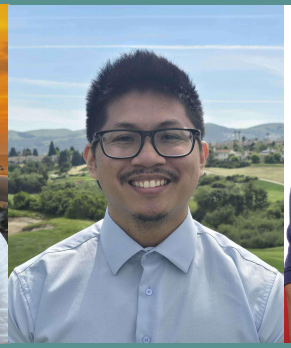
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Team Bio



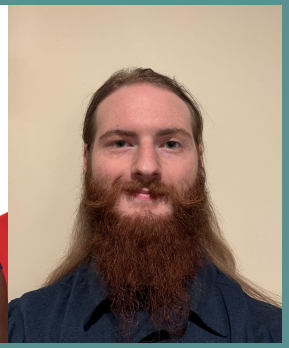
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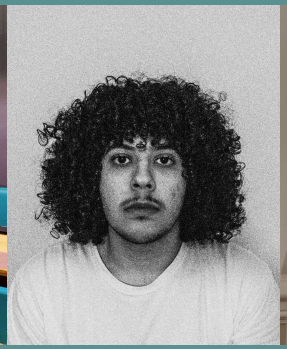
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Elevator Pitch

Classes Together creates a convenient space for informal academic discussion and streamline invitations. There are not many easy or convenient ways for students to communicate informally with their classmates outside of class. Additionally, invitations from other informal platforms are time-consuming to send and do not always stay open. Classes Together automatically enrolls students in forums for their current classes, which stay open to join for the full class duration.

High-Level Overview

Classes Together is a web-based platform that provides students with course-specific communities for informal academic discussion and collaboration. Students would sign in using their university credentials and automatically see communities for their enrolled courses, eliminating the need to manually create group chats or send invitations. These communities would allow students to ask questions, share resources, and communicate with classmates throughout the semester. The prototype will use mock enrollment data instead of a live university integration.

Problem Statement

Learning management systems (LMS) are for formal academic discussion and discourage students from asking questions due to their strict atmosphere (Lav et al., 2025). Informal instant messaging platforms provide a relaxed atmosphere, but they are not originally intended for academics or come with courses built in.

LMS only allow instructors or administrators to customize courses. While instant messengers allow user customization, popular options do not have built-in academic features.

Only LMS instructors or administrators may create invites. And while users on instant messengers may, invitations must be manually sent out and typically expire, causing students to miss invitations.

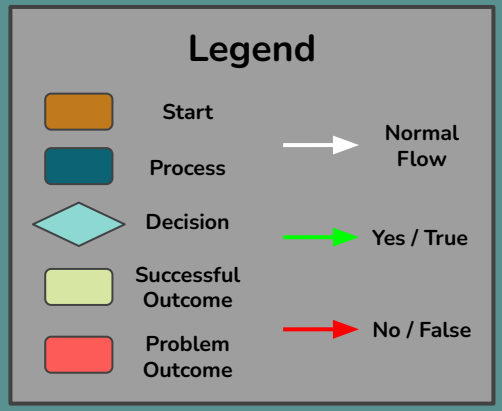
Problem Characteristics

1. Peer collaborations supports learning
    a. Students can ask each other for questions, explain difficult concepts, and/or share different perspectives outside of lectures.
2. Communication helps students feel connected
    a. Interactions with classmates can help students develop a sense of belonging within their local community.
3. Connecting with classmates can be harder online
    a. Online students have a lot fewer opportunities to meet classmates and develop relationships through casual interactions, unlike in-person interactions that students would get face-to-face.

Current Process Flow

1. Student enrolls in a course.
2. Search for an existing group chat.
    a. Discord, GroupMe, WhatsApp, Slack
    b. May need to ask your classmates if one already exists.
3. If one doesn't exist, manually create a group chat.
    a. Platform choice is dependent on who creates the group.
    b. No standardized communication platform (each course may vary).
4. Send an invitation link (could expire?) to your classmates.
5. Student who receives/finds the invitation can join.

Current Process Flowchart



Student wants to communicate with classmates outside of class

Does a class group exist?

Search for an invitation link

Does student have invitation link?

Can the student join the group?

Ask classmates for a valid invitation

Join the class group chat

Ask classmates about creating a group chat

Will somebody create a group chat?

Create the group chat on a third-party platform (e.g., Discord, GroupMe, WhatApp)

Share the invitation link with classmates

Are all enrolled students aware of the group?

Students can connect and communicate informally

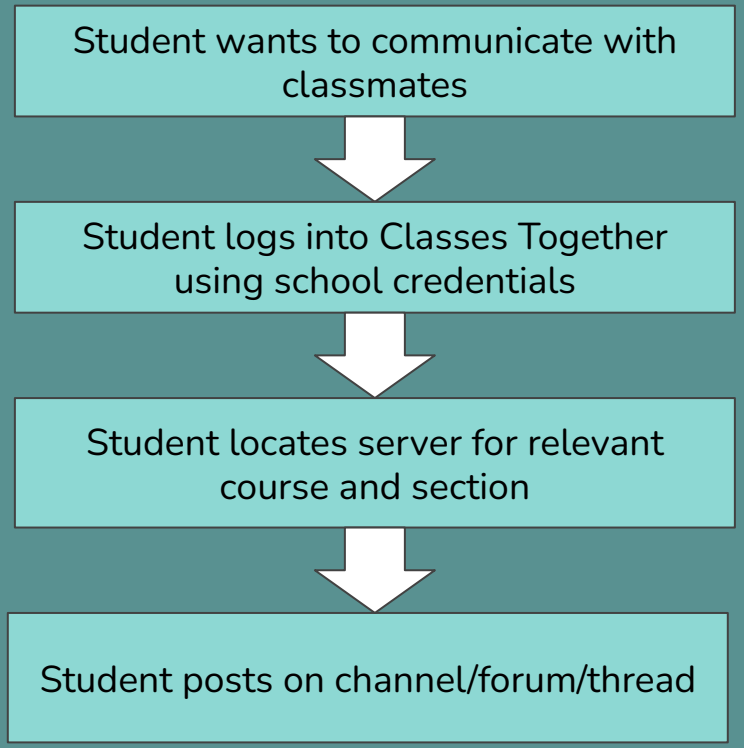
Students remain disconnected

Some students remain unaware or excluded

Solution Statement

Classes Together will eliminate the time lost by manually sending out invitations and reentry of course and academic information into a new profile. It will also provide an informal setting still academic oriented.

Solution Process Flow



What it Will Do

- Allow students to sign in to the platform.
- Display course communities based on student enrollment information.
- Provide a space for students to communicate informally with classmates.
- Organize communities by semester and then into courses.
- Simplify access to course groups without requiring students to create and distribute invitations manually.
- Allow students to create group channels within a course.

What it Will Not Do

1. It will not replace Canvas or other LMS platforms.
2. It will not be the method for instructor-to-student communication.
3. It will not require students to manually create and distribute casual course groups.
4. It will not require students to download a third-party software.
5. It will not require students to register for an additional account.
6. It will not keep course communities active indefinitely.

Competition Matrix

	Informal	Academic	Course Information Already Present	Course Invite Always Open	Course Calendar and Deadlines Already Present	Multiple Channels or Forums for One Course	Students May Modify Forum	Students May Create Invites
Canvas	No	Yes	Yes	No	Yes	Yes	No	No
Google Classroom	No	Yes	Yes	No	Yes	Yes	No	No
Discord	Yes	No	No	No	No	Yes	Yes	Yes
GroupMe	Yes	No	No	No	No	No	Yes	Yes

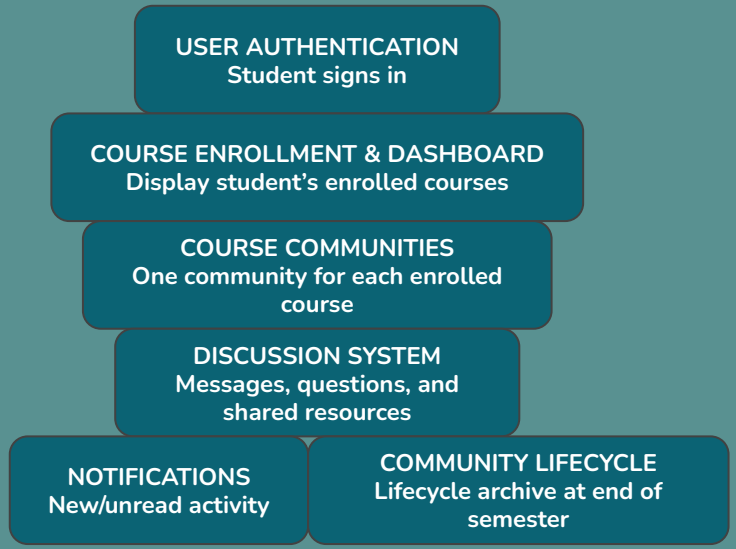
Development Tools

Integrated Development Environment (IDE)	VSCode
Version Control	Git, GitHub
Continuous Integration (CI) and Continuous Deployment (CD)	GitHub Actions and Workflows
Selected Languages (Frontend)	HTML, CSS, JavaScript
Selected Languages (Backend)	JavaScript (Node.js)
Database	PostgreSQL
Testing Frameworks	Jest (JavaScript), Playwright
Documentation Tools	JSDoc (JavaScript), GitHub README, and Wiki

Major Functional Components

1. User Authentication
2. Course Enrollment/Dashboard
    a. Displays active/inactive courses.
3. Course Communities
    a. Displays semester-specific community for that course
    b. Only students enrolled in that course may have access
4. Discussion System *(real-time messaging or forum-based..?)*
5. Notifications for unread or new activity
6. Community Lifecycle (archive/read-only when semester ends)

Major Functional Components Diagram



Risk Evaluation

Customer & End User Risks	Technical Risks	Security Risks	Legal Risks
Low popularity of this application may cause fewer users to enroll, defeating the purpose of sharing a forum with classmates when they are not present.	Real course and enrollment data may be difficult to access.	Students may access course invites they are not enrolled in.	Processing real academic emails and course information may issue takedown requests.
Probability: 5 Impact: 3	Probability: 5 Impact: 4	Probability: 3 Impact: 4	Probability: 4 Impact: 4
Students may post spam or share answers to assignments, which could negatively affect class discussions and grades.	Courses and course details cannot be determined from email alone.	By joining a class forum, students can imply each other's personal information, such as legal names and location.	K-12 forums will be subject to child protection laws.
Probability: 4 Impact: 5	Probability: 5 Impact: 4	Probability: 3 Impact: 5	Probability: 5 Impact: 3

Risk Mitigation

Customer & End User Risk Mitigations	Technical Risk Mitigations	Security Risk Mitigations	Legal Risk Mitigations
Keep course groups easy to find and use while gathering feedback from students.	Use mock enrollment data in the prototype. And in the product, allow searching for courses and adding new courses after moderator approval.	Verify course enrollment before allowing access to discussions.	Use only publicly available data or data granted by institutions.
Probability: 3 Impact: 2	Probability: 2 Impact: 1	Probability: 2 Impact: 4	Probability: 2 Impact: 1
Establish community rules and consider adding reporting and moderation features.	Allow course details to be added by users after approval or by users granted course moderator.	Restrict what personal information is public and ban sharing of certain details.	Enable content and interaction restrictions on 18- accounts.
Probability: 3 Impact: 3	Probability: 2 Impact: 2	Probability: 2 Impact: 3	Probability: 2 Impact: 1

Real World Product vs Prototype Table

Feature	Real World Product	Prototype
User Authentication	University email or SSO	Simulated login with test accounts
Course Enrollment	Automatic synchronization with enrollment records	Randomized enrollments per user
Course Dashboard	Student's actual student courses	Displays user's enrolled courses
Course Communities	Private, course discussions	View/respond to discussion threads?
Database	Stores user and course information	Test users, assigned random courses
Notifications	New discussions, replies, and mentions	Simple unread indicator

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Appendix